

Process Documentation for the Python Solvers

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This document summarizes the process implemented in the uploaded Python solvers that model a single-stage axial compressor. The files described are:

- `SingleAxialStageModel.py`
- `root_tip_axial_stage_uniform_dehalleries.py`

Common Inputs and Assumptions

Both solvers start from a set of design inputs and apply standard axial-compressor relationships to compute velocities, flow angles, and performance indicators.

Key inputs

- Rotational speed (RPM)
- Meanline radius and inlet annulus area
- Inlet flow angle (IGV angle, a_1)
- Target stage pressure ratio (not used directly in the current algebra)
- Target specific enthalpy rise, Δh
- Mass flow rate
- Target deHaller's ratio (W_2/W_1)

Derived geometry

- Annulus half-height is computed from inlet area and meanline radius.
- Hub and tip radii are computed by subtracting or adding that offset to the meanline radius.

Core kinematic and thermodynamic relations (as implemented)

- Annulus half-height:

$$\Delta r = \frac{A_{in} \cdot 10^6}{4\pi r_m}$$

(The factor 10^6 converts A_{in} in m^2 to mm^2 for consistency with r_m in mm .)

- Hub and tip radii:

$$r_{hub} = r_m - \Delta r, \quad r_{tip} = r_m + \Delta r$$

- Axial velocity (constant density assumption with $\rho = 1.225 \text{ kg/m}^3$):

$$c_z = \frac{\dot{m}}{\rho A_{in}}$$

- Tangential speed at a radius r :

$$U(r) = \left(\frac{r}{1000} \right) \left(\frac{2\pi}{60} \right) \text{RPM}$$

(Radius is converted from mm to m.)

- Stage loading (from input enthalpy rise):

$$\psi = \frac{\Delta h}{U^2}$$

- Euler work relation (used to link swirl change and enthalpy):

$$\Delta h = U (c_{02} - c_{01})$$

Solver 1: SingleAxialStageModel.py

This solver performs a meanline-only design computation for a single rotor–stator stage.

Step 1: Convert inputs and compute axial velocity

- Convert a_1 to radians.
- Compute axial velocity c_z using mass flow, density assumption, and inlet area.
- Compute the meanline tangential speed U from radius and RPM.

Formulas used

- Convert a_1 :

$$a_1 = \alpha_1 \cdot \frac{\pi}{180}$$

- Axial velocity:

$$c_z = \frac{\dot{m}}{\rho A_{in}}$$

- Meanline tangential speed:

$$U = U(r_m)$$

Step 2: Stage loading and swirl

- Compute stage loading from $\Delta h/U^2$.
- Use the Euler work relation to solve for the rotor exit relative angle b_2 .

Formulas used

- Stage loading:

$$\psi = \frac{\Delta h}{U^2}$$

- Rotor exit relative angle b_2 is computed from:

$$b_2 = \tan^{-1} \left(\frac{(\Delta h/\dot{m})/U - U}{c_z} + \tan(a_1) \right)$$

The code uses $W_{r1} = \dot{m}\Delta h$ and substitutes $\Delta h = W_{r1}/\dot{m}$.

Step 3: Inlet and exit velocity triangles

- Compute c_{01} from a_1 and c_z .
- Compute the rotor inlet relative angle b_1 .
- Compute the absolute exit angle a_2 .
- Use a deHaller's constraint to estimate the stator exit angle a_3 .

Formulas used

- Inlet tangential component:

$$c_{01} = c_z \tan(a_1)$$

- Rotor inlet relative angle:

$$b_1 = \tan^{-1} \left(\frac{U - c_{01}}{c_z} \right)$$

- Absolute exit angle:

$$a_2 = \tan^{-1} \left(\frac{U}{c_z} + \tan(b_2) \right)$$

- Stator exit angle from deHaller's ratio (midline target):

$$a_3 = \cos^{-1} \left(\frac{\cos(a_2)}{\text{deHaller}} \right)$$

If the inverse cosine is invalid, the script falls back to $a_3 = 0$.

Step 4: Derived quantities and outputs

- Compute degree of reaction from a_1 and a_2 .
- Compute W_1 , W_2 , and deHaller's ratio.
- Print angles, radii, and key performance metrics for the meanline design.

Formulas used

- Degree of reaction:

$$R = 1 - \frac{c_z}{2U} (\tan(a_2) + \tan(a_1))$$

- Rotor-relative speeds:

$$W_1 = \sqrt{c_z^2 + (U - c_{01})^2}, \quad W_2 = \sqrt{c_z^2 + (U - c_{02})^2}$$

- deHaller's ratio:

$$r = \frac{W_2}{W_1}$$

with $c_{02} = c_z \tan(a_2)$.

Solver 2: root_tip_axial_stage_uniform_dehalleries.py

This solver extends the meanline design so that the hub and tip deHaller's ratios match the midline value.

Step 1: Midline baseline

- Compute c_z , U , and the midline velocity triangles as in Solver 1.
- Compute the midline deHaller's ratio $r_{mid} = W_2/W_1$.

Formulas used

- Midline follows the same relations as Solver 1.
- Midline deHaller's ratio:

$$r_{mid} = \frac{W_{2,mid}}{W_{1,mid}}$$

Step 2: Hub matching

- Compute hub tangential speed U_{hub} .
- Compute $W_{1,hub}$ from c_z and U_{hub} .
- Enforce $W_{2,hub} = r_{mid} \cdot W_{1,hub}$ to solve for $c_{02,hub}$.
- Convert $c_{02,hub}$ to hub angles $a_{2,hub}$, $b_{2,hub}$, and $b_{1,hub}$.

Formulas used

- Hub inlet relative speed:

$$W_{1,hub} = \sqrt{c_z^2 + (U_{hub} - c_{01})^2}$$

- Enforced hub exit speed:

$$W_{2,hub} = r_{mid} W_{1,hub}$$

- Solve for $c_{02,hub}$ from:

$$W_{2,hub}^2 = c_z^2 + (U_{hub} - c_{02,hub})^2$$

The solver uses the positive root:

$$c_{02,hub} = U_{hub} - \sqrt{W_{2,hub}^2 - c_z^2}$$

- Hub angles:

$$a_{2,hub} = \tan^{-1} \left(\frac{c_{02,hub}}{c_z} \right), \quad b_{2,hub} = \tan^{-1} \left(\tan(a_{2,hub}) - \frac{U_{hub}}{c_z} \right)$$

$$b_{1,hub} = -\tan^{-1} \left(\frac{U_{hub} - c_{01}}{c_z} \right)$$

Step 3: Tip matching

- Compute tip tangential speed U_{tip} .
- Compute $W_{1,tip}$ and enforce $W_{2,tip} = r_{mid} \cdot W_{1,tip}$.
- Solve for $c_{02,tip}$ and convert to tip angles $a_{2,tip}$, $b_{2,tip}$, and $b_{1,tip}$.

Formulas used

- Tip inlet relative speed:

$$W_{1,tip} = \sqrt{c_z^2 + (U_{tip} - c_{01})^2}$$

- Enforced tip exit speed:

$$W_{2,tip} = r_{mid} W_{1,tip}$$

- Solve for $c_{02,tip}$ from:

$$W_{2,tip}^2 = c_z^2 + (U_{tip} - c_{02,tip})^2$$

with

$$c_{02,tip} = U_{tip} - \sqrt{W_{2,tip}^2 - c_z^2}$$

- Tip angles:

$$a_{2,tip} = \tan^{-1} \left(\frac{c_{02,tip}}{c_z} \right), \quad b_{2,tip} = \tan^{-1} \left(\tan(a_{2,tip}) - \frac{U_{tip}}{c_z} \right)$$

$$b_{1,tip} = -\tan^{-1} \left(\frac{U_{tip} - c_{01}}{c_z} \right)$$

Step 4: Checks and reporting

- Verify hub and tip deHaller's ratios match the midline value.
- Compute enthalpy rise and degree of reaction at hub, midline, and tip.
- Print a structured report of operating conditions, geometry, angles, and velocities.

Formulas used

- Enthalpy rise at each radius:

$$\Delta h(r) = U(r) (c_{02}(r) - c_{01})$$

- Degree of reaction at each radius:

$$R(r) = 1 - \frac{c_z}{2U(r)} (\tan(a_2(r)) + \tan(a_1))$$

- deHaller's ratio at each radius:

$$r(r) = \frac{W_2(r)}{W_1(r)}$$

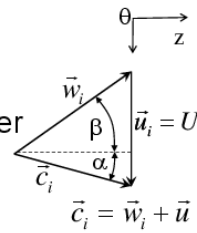
Notes and Current Limitations

- Density is assumed constant in the axial velocity calculation.
- The desired pressure ratio input is not yet used in the current computation flow.
- Both solvers treat the stage as a single meanline design with extensions to hub and tip in Solver 2.

Appendix: Velocity Triangle Reference (Page 5)



Velocity Triangles



- Rotor motion in θ direction, so ref. frame change has no effect on other directions $\Rightarrow w_{zi} = c_{zi}$ (II.13)
- Rotor vel. constant in cascade flow (like fixed r), and let $|u_i| = U$
 $\Rightarrow c_{\theta i} - w_{\theta i} = U$ (II.14) as shown here, $w_{\theta i} < 0$
- Also have general trigonometric relations

— e.g.,

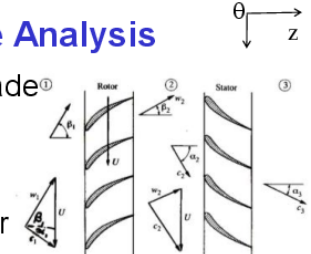
$$\begin{aligned} c_{\theta i} &= c_i \sin \alpha_i = c_{zi} \tan \alpha_i \\ w_{\theta i} &= w_i \sin \beta_i = w_{zi} \tan \beta_i \\ w_{zi} &= w_i \cos \beta_i = c_{zi} = c_i \cos \alpha_i \end{aligned}$$

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Velocity Triangle Analysis



- Apply (II.12-14) to both blade rows of compressor stage
- 1: rotor inlet
 $(13) \Rightarrow c_{\theta 1} = c_{z1} \tan \alpha_1$
- 2: between rotor and stator
 $(13,15) \Rightarrow w_{\theta 2} = c_{z2} \tan \beta_2$
 $(14) \Rightarrow c_{\theta 2} = U + w_{\theta 2}$

$c_{\theta 2} = U + c_{z2} \tan \beta_2$ (II.16)

 with (15) $\Rightarrow \tan \alpha_2 = \frac{U}{c_{z2}} + \tan \beta_2$ (II.17)
- 3: stator outlet
 $(15) \Rightarrow c_{\theta 3} = c_{z3} \tan \alpha_3$

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